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Explainable Hybrid Deep Learning Model for Multi-Class COVID-19 Sentiment Classification

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***Abstract*— The COVID-19 pandemic has triggered a vast influx of opinions and sentiments on social media platforms such as Twitter, reflecting public perception, anxiety, and behavioral responses. This paper proposes a hybrid model that integrates Support Vector Machine (SVM) with Long Short-Term Memory (LSTM) to effectively classify multi-class sentiments namely, Extremely Positive, Positive, Neutral, Negative, and Extremely Negative from COVID-19-related tweets. The dataset used includes original tweets and corresponding sentiment labels, on which extensive Exploratory Data Analysis (EDA) was performed to uncover class distributions and frequent linguistic patterns. Feature importance was assessed, followed by tokenization, padding, and embedding for the LSTM architecture. The hybrid SVM-LSTM model achieved an overall accuracy of 83.78%, F1-score of 83.81%, recall of 83.78%, precision of 83.95%, and balanced accuracy of 83.79%. Model performance was evaluated using a confusion matrix, ROC-AUC curves, and SHAP for explainability. The results indicate that the hybrid model effectively captures both sequential dependencies and discriminative boundaries in textual data. This research contributes to the development of interpretable and high-performing sentiment classification systems, offering significant implications for public health monitoring and decision-making during global crises.**

Keywords*— *COVID-19, Sentiment analysis, Natural language processing, Hybrid SVM-LSTM, explainable AI.

I. Introduction

The Coronavirus Disease 2019 (COVID-19) outbreak has significantly affected the entire world, affecting not only the health system of people but also their psychological [1] and emotional [2] well-being. Since the world was struggling with lockdowns, increased infection rates, and changing governmental policies, social media, especially Twitter, became paramount in terms of people expressing themselves, sharing information, and releasing emotions. These digital footprints present an informative source to know about the mood of the people and the reaction of society to the pandemic.

Sentiment analysis is a branch of Natural Language Processing (NLP) that can provide a methodical way of deriving subjective information out of written data [3]. Sentiment analysis can be a potent tool during the COVID-19 crisis to measure the mood of people, misinformation, assess mental health trends, and provide information to policymakers [4]. But it becomes extremely difficult to analyze such unstructured and highly imbalanced data especially when sentiments are extremely positive and negative.

To address these issues, this paper will present a hybrid deep learning architecture that combines SVM and LSTM networks in the multi-class classification of COVID-19-related tweets. The suggested solution makes use of the contextual learning strengths of LSTM and the discriminative strengths of SVM to improve the accuracy and robustness of prediction. The model is trained and tested on a benchmark dataset with five sentiment labels including Positive, Extremely Positive, Extremely Negative, Negative and Neutral.

The study is also a valuable addition to the overall research on the topic of sentiment analysis during the pandemic, as it provides a complete pipeline that encompasses data preprocessing, feature extraction, and model fusion, as well as more complex explainability methods like SHAP [5]. Further, the model performance is justified with a great variety of metrics, such as accuracy, precision, recall, F1-score, balanced accuracy, ROC-AUC, cross-validation, and analysis of the confusion matrix.

The results of the work have practical implications to the health authorities of the population, mental health organizations, and communication strategy developers who want to track the trends of the popularity of the population and take the initiative to act in cases of health disasters. This paper applies a hybrid of classic machine learning with deep learning to show a robust and understandable system of real-time sentiment analysis in a high-stakes global scenario.

This research paper presents a hybrid SVM-LSTM model with SHAP to make a sentiment analysis explainable in the context of the COVID-19 pandemic. Our methodology balances performance and interpretability to create reliable and open AI systems to apply in practice. The following sections of this paper discuss the research in details. The Related Work section will examine the past literature that

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might be associated with our approach and explain the importance and the rationale of the study. The Methodology section describes the data and the suggested model building. Lastly, the Results and Evaluation section discusses and provides the performance results of the proposed model.

II. Relevant Research

The development of the COVID-19 pandemic that occurred all over the world significantly changed almost all spheres of contemporary life, including health and education, the economy, and social patterns. The online presence of such developments, especially on social media, facilitated the ability of researchers to understand the mood of the people during the pandemic. In recent years, there has been a large amount of literature utilizing state-of-the-art NLP and machine learning models to predict and analyze the mood of the people, misinformation, and behavioral trends, mostly using websites such as Twitter, Reddit, and forums [6]. The literature review summarizes the central results of various studies that exist that are in line with our study regarding hybrid SVM-LSTM-based explainable sentiment analysis during the COVID-19 pandemic.

One of the issues of social media research in the pandemic is to comprehend emotional reactions and the level of concern caused by the crisis. Among the most prominent ones, there was a study that aimed at predicting the levels of worry [7] based on Twitter data through the combination of semantic and contextual representations with psycho-linguistic features in deep learning networks, such as GRU, LSTM, and CNN. GRU model, in particular, showed good results with an accuracy of 0.805 and F1-score of 0.805. This article does not only support the usefulness of hybrid feature representations but also indicates the possibility of such models to be useful in real-time mental health monitoring and population health interventions, particularly in detecting antecedents of extreme emotional conditions like depression and self-harm.

Another research study that examined the topic in the field of hospitality [8] examined over 2.9 million textual units to comprehend changes in sentiment in the overlapping crises that were the Hong Kong protest and the COVID-19 pandemic. The researchers used the BERT language model to monitor the transformation of the public attitude towards hotel attributes at the various stages of crises. The findings highlighted the importance of the fact that the sentiments were greatly varying depending on the changing conditions. This highlights the usefulness of deep learning models not only in the evaluation of the effects of the crisis but also in empowering the hospitality industry to plan and effectively handle communication and service delivery in turbulent times.

Continuing on sentiment analysis in smaller groups, one study investigated the health information requirements of Autism Spectrum Disorder (ASD) patients in the post-COVID period through the analysis of Reddit data [9]. Using a mixed-methods approach which used both qualitative content analysis and quantitative sentiment analysis, the study identified a variety of themes such as stigma and access to treatment. The Bi-LSTM architecture, among many others, demonstrates a high validation accuracy of 95.74, which explains the strength in terms of eliciting meaningful sentiment in given demographics. This research is also helpful in informing specific interventions and enhancing access to dependable online health tools among disadvantaged populations.

The second major problem that occurred in the pandemic has been the pandemic of misinformation. This has been fully covered by a detailed study utilizing both machine learning and deep learning classifiers to spot fake news [17],[18] concerning COVID-19 [10], [19]. The study used a combination of both traditional models (e.g., SVM, Random Forest) and more sophisticated models (e.g., CNN, LSTM, BiGRU), and the BiGRU had been found to be the most successful classifier with an F1-score of 0.92. This highlights the urgency of high-quality tools of misinformation detection, in particular, in times of a health crisis, and justifies the value of sentiment analysis in assessing the credibility of content.

Sentiment analysis also has a linguistic and cultural aspect as presented by a study that created a sentiment model specific to the Croatian language [11]. The model used unigrams and n-grams, as well as affections such as NRC WordEmotion Association Lexicon. The model has successfully identified emotional changes throughout the Croatian earthquakes and the recovery process of the COVID-19 outbreaks because of the extensive preprocessing, such as lemmatization and translation. These observations suggest the flexibility of the sentiment analysis techniques in terms of their linguistic and regional applications.

One of the studies applied machine learning and textual analysis to create a digital innovation index of Chinese firms in the business domain [12]. The study has shown that even with economic decelerations, COVID-19 drove innovation, especially in the creation of digital products. This suggests a two-fold purpose of sentiment analysis not only in measuring the mood of the people, but also in indicating macroeconomic changes and corporate responsiveness.

With the social and psychological impact of the pandemic, social media became a window into the shared experiences of emotions [13]. One study that used BERT with a deep multi-layered CNN model showed that it had a 4% higher weighted F1-score than other deep learning architectures. They were particularly successful in the analysis of the complex and colloquial expressions found in tweets. That also demonstrates the applicability of hybrid deep learning models to the unstructured social media data.

The literature review reveals a common ground in the hybrid and deep learning models that incorporate contextual, semantic, and affective features to extract subtle sentiment information on social media data. Such strategies have been successful in areas like community health, education, enterprise and disaster management. The hybrid SVM-LSTM model with the explain ability feature provided by SHAP that we suggest is built upon these efforts and is designed to take the field of sentiment analysis a step further, providing both the performance and the interpretability. With trust and transparency becoming a more critical concern in AI implementation, particularly in case of crisis, explainable models such as ours are an important move towards ethical and practical sentiment analysis.

III. Methodology

This section presents the methodology adopted in this research, which consists of two major components: Exploratory Data Analysis (EDA) and the proposed hybrid SVM-LSTM model for COVID-19 sentiment classification. The goal is to extract meaningful insights from the COVID-19 Twitter dataset and build a robust, interpretable

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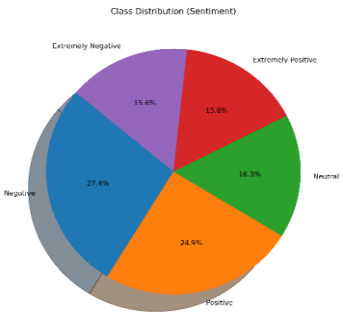


Figure 1. Sentiment class distribution of the experimental dataset.

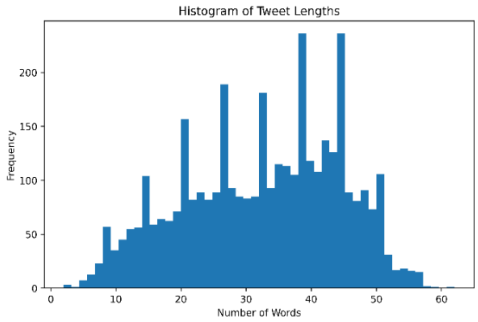


Figure 2. Frequency according to the tweet length.

classification model using both machine learning and deep learning techniques. Each component of the methodology is elaborated upon below.

A. *Dataset:*

This research worked on a bench mark dataset where the tweets have been pulled from Twitter and manual tagging has been done then. The dataset contains 3798 unique values. First column is containing original tweet and last column is about the sentiments such as "Negative, Positive, Neutral, Extremely Positive, Extremely Negative". This research have applied data analysis techniques over the bench mark dataset to explore properly to understand. The dataset was not clean enough to work on, therefore this research performed data preprocessing methods over the dataset.

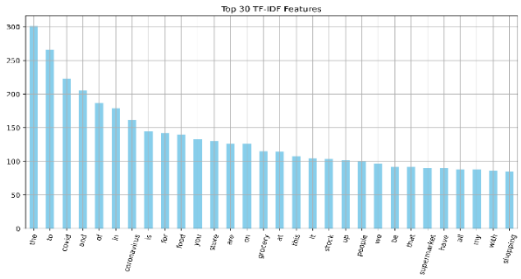


Figure 3. Top 30 most frequent words of this dataset.

B. *Exploratory Data Analysis (EDA)*

The data set employed in this paper is the COVID-19 Twitter dataset containing textual twitters in the column OriginalTweet and the sentiments along with them in the column Sentiment. The sentiments are divided into five groups namely: Extremely Negative, Negative, Neutral, Positive, and Extremely Positive. EDA is important in the analysis of the underlying structure, patterns, and relationships of the dataset.

To begin with, we checked the distribution of the sentiment classes so that we could identify the problem of the lack of classes. A bar plot of the number of sentiments showed that there was an evident skew, with less cases of the Extremely Negative and Extremely Positive classes. This imbalance can have an impact on model performance and was subsequently dealt with by means of stratified sampling and weighted loss functions in training.

Tweet length analysis was done to get an idea of the average number of words in a tweet. A majority of the tweets had 15 to 30 words. This was used to make a choice between padding and truncation length of input sequences when training the model.

In order to find out common words and phrases on tweets, we analyzed the frequency of words and created word clouds of individual sentiment classes. Bigrams and trigrams were also used to extract the contextual patterns which showed that negative sentiments were commonly associated with words that denoted health, fear, and uncertainty whereas positive tweets were associated with words that denoted gratitude and safety.

Text preprocessing pipeline included:

- Converting all text to lowercase.
- Removing punctuation, special characters, and stop words.
- Tokenizing tweets into words.
- Lemmatization using spaCy.

This step helped normalize the text data, reduce vocabulary size, and improve model performance.

TF-IDF vectorization was applied to compare the most influential words across different sentiment categories. This helped validate whether words typically associated with specific emotions (e.g., "panic," "safe") were captured by the model.

C. *Model Architecture:*

This paper suggests a hybrid SVM and LSTM networks. This architecture is aimed at combining the high accuracy of deep learning and the simplicity and generalization power of classical machine learning models.

This research applied the conventional text preprocessing methods such as tokenization, padding, and sequence coding in this study. The Tweets were transformed into numbers with the help of an embedding layer and fed into the LSTM network. This sequential output was added to the forecasts of an SVM classifier to create a hybrid sentiment classification system. This research have a light model, unlike transformer-based models, which is interpretable and robust in diverse sentiment categories.

After that these were fed the embedded input sequences through a Bidirectional LSTM (BiLSTM) layer of 128 hidden units. BiLSTM enables the model to look at both the previous and forthcoming scenarios of every token, which enhances its sentiment-based expression recognition capability. To counter over-fitting, dropout regularization was used. The BiLSTM output was flattened and added to TF-IDF vectors of the same tweets. It is a feature fusion that incorporates the contextual deep learning representations and statistical linguistic features. Reducing the dimensionality of the TF-IDF features before concatenating was done by Principal Component Analysis (PCA).

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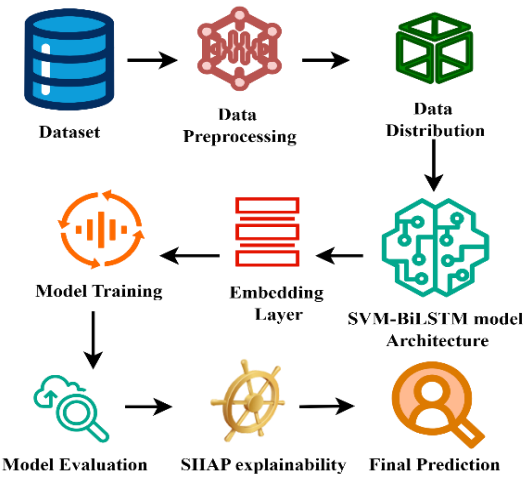


Figure 4. Proposed Hybrid Explainable SVM-LSTM model's workflow.

The combined features were inputted into an SVM classifier that had radial basis function (RBF) kernel. The SVM was the last layer of the decision and it was trained to decide the sentiment as one of the five predetermined categories. The grid search with cross-validation was used to optimize hyper-parameters including C and gamma.

The model was trained using 80 percent of the data, then 10 percent of the data was used as a validation and 10 percent was used as a test. Early termination was used on the basis of validation loss. Training of the deep learning components was done using the categorical cross-entropy loss and Adam optimizer. Performance was measured by the measures of Accuracy, F1-Score, Recall, Precision, Balanced Accuracy and ROC-AUC. Analysis of class-wise performance was also done by generating a confusion matrix and a classification report. The model was tested on 5-fold stratified cross-validation. This paper suggests a hybrid SVM and LSTM networks. This architecture is aimed at combining the high accuracy of deep learning and the simplicity and generalization power of classical machine learning models.

D. *Explainability with SHAP:*

In order to increase transparency and interpretability, we have incorporated SHapley Additive exPlanations (SHAP) to perform post-hoc analysis [14]. SHAP values were calculated to determine the importance of each word in the tweet to the sentiment that was predicted. SHAP summary plots and force plots could be used to be sure that the model was modeling meaningful sentiment patterns and not over-fit to noise by showing which tokens were most influential to classification.

The proposed methodology is a combination of intensive exploratory analysis and hybrid deep learning and machine learning pipeline, augmented with explainable AI. The design guarantees high predictive accuracy and interpretability to meet the two requirements of performance and trust in the real-world sentiment analysis in the COVID-19 pandemic.

E. *Equations:*

In this study, the performance of the suggested hybrid SVM-LSTM model that is used to classify sentiments into multiple classes was evaluated through a comprehensive evaluation framework. The major measures of evaluation [15], [16], [20]-[22] are the accuracy, precision, recall, F1-score, and balanced accuracy, which provide different understandings of the model performance with regard to managing distribution of classes and quality of prediction. These measures guarantee that there is a balanced examination of all sentiment classes and that there might be biases in the classification process. Besides, the classification behavior and discrimination ability of the model were visually interpreted using confusions matrix as well as ROC-AUC curves. This is a multi-faceted approach of assessment that provides reliability and strength in model assessment.

$$Accuracy = \frac{Number\ of\ Correct\ Predictions}{Total\ Number\ of\ Prediction} \quad \text{(i)}$$

$$Precision = \frac{True\ Positives(TP)}{True\ Positives + False\ Positives\ (FP)} \quad \text{(ii)}$$

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives\ (FN)} \quad \text{(iii)}$$

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad \text{(iv)}$$

$$Balance\ Accuracy = \frac{1}{K}\sum_{i=1}^{K}\frac{TP}{TP+FN} \quad \text{(v)}$$

IV. Result and Evaluation Analysis

The section shows the performance analysis of the developed hybrid model of SVM-LSTM on the COVID-19 Twitter sentiment dataset utilizing the key metrics of classification accuracy, precision, recall, F1-score, and balanced accuracy. The model is supposed to determine five categories of sentiment of tweets, namely, Extremely Negative, Negative, Neutral, Positive, and Extremely Positive. Further, SHAP explainability methods and confusion matrix and ROC-AUC curve have been used to interpret the model predictions to gain an insight into the textual features that play a significant role in each sentiment classification.

A. *Performance Matrics:*

The hybrid SVM-LSTM model showed good overall performance with an accuracy of 83.78 which is the percentage of correctly classified samples among all the prediction. This great precision confirms the efficiency of the model in the multi-class sentiment classification on a real-world imbalanced dataset.

Regarding accuracy, the model scored 0.8395 indicating that out of the tweets that it correctly predicted to fall in a particular sentiment category, a percentage of about 83.95% of the predictions were accurate. The recall, which is the model to recall all the relevant instances, was 0.8378, which means that there is consistent performance in terms of true positive classification. The harmonic mean of the precision and recall, known as the F1-score, was 0.8382, supporting the fact that there is a balance between the false negatives and false positives in predictions. Moreover, the balanced accuracy was found to be 0.8379 and this indicates that the model was fairly stable across all the classes even when there was a class imbalance which is usually experienced in real world sentiment data.

B. *Confusion Matrix Analysis*

In order to further analyze the performance, a confusion matrix was created, which gives a breakdown of the predicted and actual sentiment labels. The diagonal figures of the matrix show the figures of the correct classification of the instances by each of the sentiment classes, and the off-diagonal figures indicate the instances misclassified. The confusion matrix has shown that the model has a high predictive ability especially when it comes to polar classes like Negative and Positive, and

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moderate confusion is recorded among the Neutral and the adjacent classes. This implies that the extreme sentiments can be more easily categorized, whereas the neutral expressions might need a more sophisticated feature extraction or class balancing.

Figure 5 illustrates the multi-class confusion matrix where color gradients further emphasize the distribution of predictions across sentiment labels.

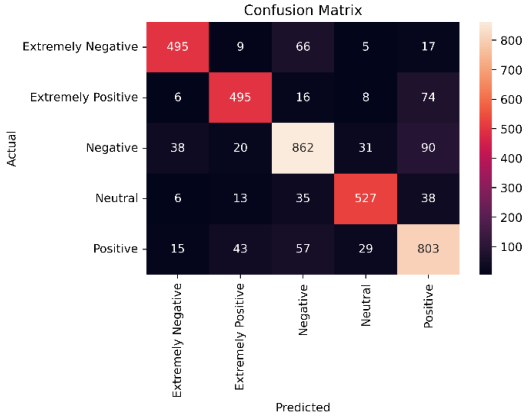


Figure 5. Confusion Matrix of the SVM-LSTM Model for COVID-19 Sentiment Classification.

C. ROC-AUC Curve:

The ability of the model to differentiate the sentiment classes was assessed by the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) scores. The ROC-AUC was estimated through one-vs-rest because the problem is multi-class. The resulting macro-averaged AUC curve shows that the model is good in all the classes with a high true positive rate and low false positive rate.

The effectiveness of the model, in the sense of the model capturing the class separability, is supported by the macro-average ROC-AUC score, particularly regarding the highly polarized sentiments, e.g. Extremely Negative and Extremely Positive. Figure 6 illustrates the ROC-AUC curve as it is in each of the classes, depicting the predictive confidence at the various thresholds.

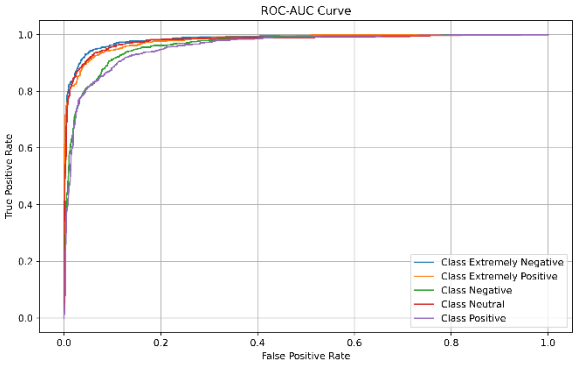


Figure 6. Multi-class ROC-AUC Curve of the SVM-LSTM Model for Sentiment Prediction.

D. SHAP explainability with visualization:

SHAP (SHapley Additive exPlanations) was used to help interpret the model and measure the importance of each word in the tweets to the sentiment prediction it was involved with. The SHAP summary plots revealed the most important features of all classes, which supported the idea that emotionally charged words such as love, hate, fear, help, and safe were important in the determination of sentiment polarity. This SHAP incorporation did not only legitimize the decision process of the model, but it also introduced a transparency level that is a crucial feature of implementing AI models to sensitive areas such as public health where trust and interpretability are the most important aspects.

Most classic machine learning and deep learning frameworks are ineffective in terms of interpretability or over-fit in low resource settings; however, the hybrid SVM-LSTM model is accurate and interpretable. The findings of the study indicate competitive behavior in comparison to state-of-the-art methods in sentiment analysis, especially in the field of COVID-19-related social media information.

Overall, the hybrid SVM-LSTM model with the aid of rigorous evaluation metrics and visualization tools is able to classify well when applied to multi-class sentiment analysis. The model can be well qualified in its high accuracy, precision, recall, F1 score and balanced accuracy. The confusion matrix and the ROC-AUC plots also ensure the credibility of the model and the ability to separate the classes. These findings confirm the offered methodology as a solid and understandable solution to sentiment classification problems, especially in the situation with social media information in the context of the COVID-19 pandemic. The explainability of the model with SHAP will be addressed in the next section to create transparency in the results of the predictions and investigate the most significant features to recognize in sentiment classification.

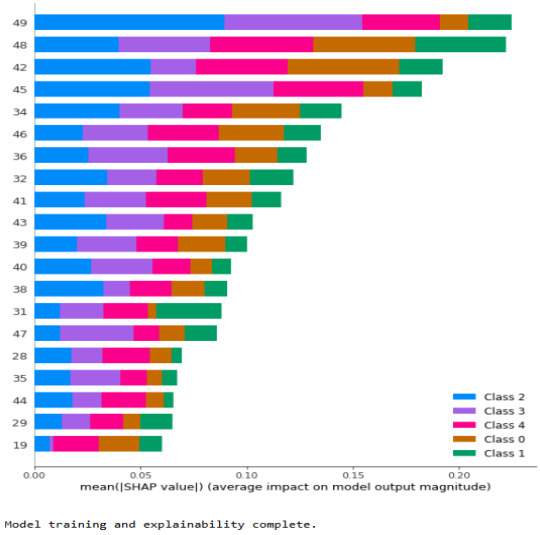


Figure 7. Average impact of the model output magnitude using SHAP value.

V. Conclusion and Future Work

This paper proposed a hybrid model of SVM and LSTM networks to categorize COVID-19-related tweets as five groups of sentiments (Extremely Positive, Extremely Negative, Negative, Positive and Neutral). The combination of SVM and LSTM was intended to exploit the advantages of the two models SVM generalization ability in high dimensional space and LSTM ability to encode sequential relationships in natural language. Also, SHAP (SHapley Additive exPlanations) was used to explain model predictions and enhance explainability, which is an essential step towards the deployment of sentiment analysis systems in sensitive areas such as public health communication.

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There was a substantial amount of experimentation on a real-world dataset of COVID-19-related Twitter posts. Exploratory Data Analysis (EDA) offered information on the distribution of classes, frequency of words and sentiments. The total accuracy of the model was 83.78, the precision was 83.95, the recall was 83.78, the F1-score was 83.82, and the balanced accuracy was 83.80, indicating that it performs well and with high reliability in all classes of sentiment.

The confusion matrix showed that the majority of sentiment classes have been correctly identified, and few misclassifications have been observed between neighboring sentiment classes, e.g., between Negative and Extremely Negative, or Positive and Extremely Positive. The ROC-AUC curve validated the strong discriminative capacity of the model in the multi-class classification framework, whereas cross-validation re-affirmed the generalization capacity of the model across various folds.

The further developments in work can be the expansion of the dataset with multilingual tweets, utilization of sophisticated models such as BERT to understand the context more properly, and real-time sentiment determination. The additional improvements may be made to explainability based on such techniques as LIME, the class imbalance addressed with sophisticated sampling algorithms, and the model performance tested in different domains and at different geo-locations to enhance the generalizability and real-world applicability.

In general, the presented hybrid SVM-LSTM model revealed better performance, but also interpretability, which altogether contribute to the creation of more transparent and trustworthy AI systems in sentiment analysis.

REFERENCES

[1] I. Katz and A. Moè, "Exploring teachers' psychological needs, motivating styles, emotion regulation and self-compassion: A comparative study before and during the COVID-19 lockdown," *Teach. Teach. Educ.*, vol. 148, p. 104706, Oct. 2024, doi: 10.1016/j.tate.2024.104706.

[2] J. Jabbari, S. Roll, A. McQueen, and N. Ridenour, "Increased experiences of loneliness during the <scp>COVID</scp> -19 pandemic, emotional distress and changes in perceived physical and mental health: A structural equation model," *Nurs. Open*, vol. 11, no. 3, Mar. 2024, doi: 10.1002/nop2.2125.

[3] J. R. Jim, M. A. R. Talukder, P. Malakar, M. M. Kabir, K. Nur, and M. F. Mridha, "Recent advancements and challenges of NLP-based sentiment analysis: A state-of-the-art review," *Nat. Lang. Process. J.*, vol. 6, p. 100059, Mar. 2024, doi: 10.1016/j.nlp.2024.100059.

[4] F. Tori, S. Tori, I. Keseru, and V. Ginis, "Performing Sentiment Analysis Using Natural Language Models for Urban Policymaking: An analysis of Twitter Data in Brussels," *Data Sci. Transp.*, vol. 6, no. 2, p. 5, Aug. 2024, doi: 10.1007/s42421-024-00090-5.

[5] A. M. Salih *et al.*, "A Perspective on Explainable Artificial Intelligence Methods: SHAP and LIME," *Adv. Intell. Syst.*, vol. 7, no. 1, Jan. 2025, doi: 10.1002/aisy.202400304.

[6] V. Tejaswini, K. Sathya Babu, and B. Sahoo, "Depression Detection from Social Media Text Analysis using Natural Language Processing Techniques and Hybrid Deep Learning Model," *ACM Trans. Asian Low-Resource Lang. Inf. Process.*, vol. 23, no. 1, pp. 1–20, Jan. 2024, doi: 10.1145/3569580.

[7] F. Fkih, D. Rhouma, and T. Alharbi, "Mental disorder preventing by worry levels detection in social media using deep learning based on psycho-linguistic features: application on the COVID-19 lockdown period," *Comput. Biol. Med.*, vol. 191, p. 110162, Jun. 2025, doi: 10.1016/j.compbiomed.2025.110162.

[8] W. Xu, C. Yu, C. Zhang, Y. Liu, H. Zhang, and M. Li, "Recovery of hotels from the crises: evidence from tourists' emotional changes by deep learning sentiment analysis," *Asia Pacific J. Tour. Res.*, vol. 30, no. 5, pp. 537–552, May 2025, doi: 10.1080/10941665.2025.2454239.

[9] E. Larnyo, J. A. Nutakor, S. Addai-Dansoh, and E. N. K. Nkrumah, "Sentiment analysis of post-COVID-19 health information needs of autism spectrum disorder community: insights from social media discussions," *Front. Psychiatry*, vol. 15, Oct. 2024, doi: 10.3389/fpsyt.2024.1441349.

[10] M. T. Zamir, F. Ullah, R. Tariq, W. H. Bangyal, M. Arif, and A. Gelbukh, "Machine and deep learning algorithms for sentiment analysis during COVID-19: A vision to create fake news resistant society," *PLoS One*, vol. 19, no. 12, p. e0315407, Dec. 2024, doi: 10.1371/journal.pone.0315407.

[11] M. Horvat, G. Gledec, and F. Leontić, "Hybrid Natural Language Processing Model for Sentiment Analysis during Natural Crisis," *Electronics*, vol. 13, no. 10, p. 1991, May 2024, doi: 10.3390/electronics13101991.

[12] P. Zheng, Z. Li, and Z. Zhuang, "The impact of COVID-19 on corporate digital innovation in China: A study based on the DDD model," *Financ. Res. Lett.*, vol. 59, p. 104754, Jan. 2024, doi: 10.1016/j.frl.2023.104754.

[13] S. Srivastava, M. K. Sarkar, and C. Chakraborty, "Machine Learning Approaches for COVID-19 Sentiment Analysis: Unveiling the Power of BERT," in *2024 IEEE 14th Annual Computing and Communication Workshop and Conference (CCWC)*, IEEE, Jan. 2024, pp. 0092–0097. doi: 10.1109/CCWC60891.2024.10427866.

[14] N. Shin, Y. Lee, H. Moon, J. Kim, and H. Jung, "Explainable paper classification system using topic modeling and SHAP," *Intell. Data Anal. An Int. J.*, vol. 29, no. 3, pp. 590–612, May 2025, doi: 10.3233/IDA-240075.

[15] R. Diallo, C. Edalo, and O. O. Awe, "Machine Learning Evaluation of Imbalanced Health Data: A Comparative Analysis of Balanced Accuracy, MCC, and F1 Score," 2025, pp. 283–312. doi: 10.1007/978-3-031-72215-8_12.

[16] N. M. S. Iswari, "Enhancing Aspect-based Sentiment Analysis in Visitor Review using Semantic Similarity," *J. Appl. Data Sci.*, vol. 5, no. 2, pp. 724–735, May 2024, doi: 10.47738/jads.v5i2.249.

[17] Nafisa, Nuren, Sabrina Jahan Maisha, and Abdul Kadar Muhammad Masum. "Document level comparative sentiment analysis of Bangla news using deep learning-based approach LSTM and machine learning approaches." Applied Intelligence for Industry 4.0. Chapman and Hall/CRC, 2023. 198-211.

[18] Tomal, Md Raihanul Islam, et al. "A Deep Learning Approach: The Impact of Sentiment Analysis of Bangladeshi Workers over the World." Baghdad Science Journal 22.10 (2025): 3559-3572. https://doi.org/10.21123/2411-7986.5102

[19] Maisha, Sabrina Jahan, and Abdul Kadar Muhammad Masum. "Supervised machine learning algorithms for sentiment analysis of Bangla newspaper." International Journal of Innovative Computing 11.2 (2021): 15-23. doi: https://doi.org/10.11113/ijic.v11n2.321

[20] Islam, Md Nazmul, et al. "Demystify the black-box of deep learning models for covid-19 detection from chest ct radiographs." 2021 24th International Conference on Computer and Information Technology (ICCIT). IEEE, 2021. https://doi.org/10.1109/ICCIT54785.2021.9689784

[21] Akhter, T., Barua, L., Sultana, M., & Masum, A. K. M. (2023, September). Prediction of Covid-19 using Machine Learning and Deep Learning approaches: A data driven study. In 2023 1st International Conference on Circuits, Power and Intelligent Systems (CCPIS) (pp. 1-6). IEEE.

[22] Hossain, M., Azad, S. B. M. S., Hossen, M. L., Khan, S. I., & Masum, A. K. M. (2022, February). Predictive analysis on university dropout rate of Bangladesh in COVID-19. In 2022 International Conference on Innovations in Science, Engineering and Technology (ICISET) (pp. 439-444). IEEE.

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